

Solve the radical equation, and check all proposed solutions

1. $\sqrt{x+5} = 7$

$$\begin{aligned} (\sqrt{x+5})^2 &= 7^2 \\ x+5 &= 49 \\ x &= 44 \checkmark \end{aligned}$$

Solve and check the equation.

2. $x^{3/2} = 8$

$$\begin{aligned} (x^{3/2})^{2/3} &= 8^{2/3} \\ x &= \sqrt[3]{64} \\ x &= 4 \checkmark \end{aligned}$$

3. $(x+6)^{5/2} - 5 = 3$

$$\begin{aligned} (x+6)^{5/2} &= 8 \\ ((x+6)^{5/2})^{2/5} &= 8^{2/5} = (2^3)^{2/5} = 2^{6/5} = 2^1 \cdot 2^{1/5} \\ x+6 &= 2\sqrt[5]{2} \\ x &= -6 + 2\sqrt[5]{2} (\approx -3.7026) \checkmark \end{aligned}$$

Solve the absolute value equation or indicate that the equation has no solution.

4. $|x+6| = 7$

$$\begin{aligned} |x+6| &= 7 \\ x+6 &= 7 \quad \text{or} \quad x+6 = -7 \\ \Rightarrow x &= 1 \checkmark \quad \Rightarrow x = -13 \checkmark \end{aligned}$$

5. $3|x+2| + 4 = 2$

$$\begin{aligned} 3|x+2| &= -2 \\ |x+2| &= -2/3 \\ |x+2| \geq 0 &\quad \text{So there is NO solution} \end{aligned}$$

Solve the polynomial equation by factoring and then using the zero product principle.

6. $x^3 + 8x^2 - x - 8 = 0$

$$\begin{aligned} x^2(x+8) - 1(x+8) &= 0 \\ (x^2-1)(x+8) &= 0 \\ (x+1)(x-1)(x+8) &= 0 \\ \Rightarrow x &= -1, 1, -8 \end{aligned}$$

Determine if the given function is even, odd, or neither.

7. $x^4 + 2x^2$.

It is even. Since the degree of each term is even. Also $f(-x) = (-x)^4 + 2(-x)^2 = x^4 + 2x^2 = f(x)$.

Give the domain and range of the relation, and determine if it is a function. [2 pts each]

8. $\{(6, 7), (10, -8), (9, 6), (1, 2)\}$

a. Domain:

$$\{6, 10, 9, 1\}$$

b. Range:

$$\{7, -8, 6, 2\}$$

c. Function?

Yes

9. $\{(5, -4), (1, -3), (1, 0), (10, -3), (26, -3)\}$

a. Domain:

$$\{5, 1, 10, 26\}$$

b. Range:

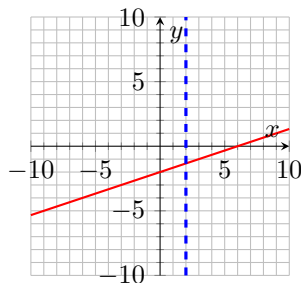
$$\{-4, -3, 0\}$$

c. Function?

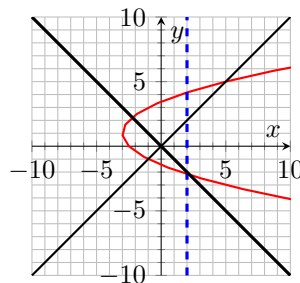
No, since 1 is paired with both -3 and 0 .

Use the vertical line test to determine whether or not the graph is a graph in which y is a function of x . [2 points each]

10. a. YES it is a function. A vertical line will pass through the curve only once.



- b. No it is not a function by the vertical line test.



Use the provided graph to complete the problem.

11. [3 points each]

a. Determine the x -intercept(s) of $f(x)$.

$$\{-5, 2\}$$

b. Determine the y -intercept(s) of $f(x)$.

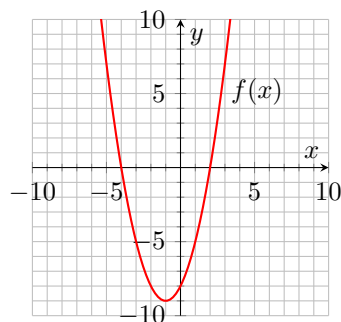
$$\{8\}$$

c. Determine the domain of $f(x)$.

$$(-\infty, \infty)$$

d. Determine the range of $f(x)$.

$$[-9, \infty)$$



Evaluate the function at the given value of the independent variable and simplify.

12. $f(x) = -5x + 1$; $f(-3)$

$$\begin{aligned} f(-3) &= -5(-3) + 1 \\ &= 16 \end{aligned}$$

13. $f(x) = x^2 - x$; $f(x - 2)$

$$\begin{aligned} f(x - 2) &= (x - 2)^2 - (x - 2) \\ &= x^2 - 4x + 4 - x + 2 \\ &= x^2 - 5x + 6 \end{aligned}$$

Find the slope of the line that goes through the given points.

14. $(2, -3)$ and $(-7, 8)$.

$$m = \frac{-3 - (8)}{2 - (-7)} = \frac{-11}{9} = -\frac{11}{9}$$

Use the given conditions to write an equation for the line in the indicated form.

15. [8 points] Passing through $(5, 8)$ with a slope of 3.

a. Point-slope form. $m = 3$

$$y - 8 = 3(x - 5)$$

b. Slope intercept form.

$$y - 8 = 3(x - 5)$$

$$y - 8 = 3x - 15$$

$$y = 3x - 7$$

16. [8 points] Passing through $(3, 8)$ and perpendicular to the line $y = -\frac{1}{3}x + 7$.

c. Point-slope form. $m = 3$ since $\frac{3}{1}$ is the negative reciprocal of $-\frac{1}{3}$.

$$y - 8 = 3(x - 3)$$

d. Slope intercept form.

$$y - 8 = 3(x - 3)$$

$$y - 8 = 3x - 9$$

$$y = 3x - 1$$